B.TECH 1ST YEAR • 1ST SEMESTER

Engineering Mathematics

Unit IV – Partial Differentiation & Its Applications

Subject: Engineering Mathematics (MA101)

Unit: IV | Level: Beginner-Friendly, Exam-Oriented

Sections: 16 | Solved Examples: 45+ | Exercise Problems: 50

Every step explained • No shortcuts • Zero confusion

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Preface & How to Use This Module

Welcome to the most thorough study guide for **Engineering Mathematics — Unit IV: Partial Differentiation and Its Applications**. This module has been designed keeping in mind a student who is seeing multivariable calculus for the very first time. No step is skipped; no concept is left unexplained.

> 📚 **Philosophy of This Module**
>
> Most textbooks show you *what* to do. This module shows you *why* every single step is done, *what* each symbol means, and *how* to think about each problem from scratch. Even if you have never done any calculus before today, you can follow every example here.

Structure of Each Section:

- **Concept Block:** Clear definition in plain English first, then mathematical language.
- **Intuition Block:** Why does this concept exist? What problem does it solve?
- **Formula Block:** Exact formula written cleanly.
- **Step-by-Step Examples:** Every example uses the 11-step framework.
- **Common Mistakes:** What students typically get wrong.
- **Summary:** Key takeaway in one or two lines.

> ⚠ **Exam Alert:** This entire unit consistently appears in B.Tech Semester examinations for 25–30 marks. Mastering this unit alone can significantly improve your overall grade. Pay special attention to Jacobians, Lagrange Multipliers, and Maxima-Minima problems.

§1 — Introduction to Multivariable Functions

1.1 What is a Function of One Variable? (Quick Recap)

Before diving into multivariable functions, let us quickly recall what we already know. A function of **one variable** is a rule that assigns exactly one output value to each input value.

DEFINITION — SINGLE VARIABLE FUNCTION

A function $y = f(x)$ takes a single input x and produces a single output y. For every value of x, there is exactly one value of y.

Examples: $f(x) = x^2$, $f(x) = \sin(x)$, $f(x) = e^x$

These functions have only one "dial" to turn — the variable x. If we change x, the output y changes accordingly. The graph is a curve in a flat 2D plane.

1.2 What is a Function of Two Variables?

In engineering and science, most real quantities depend on **more than one variable** simultaneously. Consider:

- The temperature at a point on a metal plate depends on its **x-coordinate and y-coordinate**.
- The pressure of a gas depends on both its **volume V and temperature T**.
- The profit of a company depends on both **quantity produced and price per unit**.

DEFINITION — FUNCTION OF TWO VARIABLES

A function $z = f(x, y)$ is a rule that assigns a unique real number z (output) to every ordered pair (x, y) in its domain.

Here: x and y are the **independent variables** (inputs), and z is the **dependent variable** (output).

Think of it this way: a function of one variable is like a vending machine with one button. A function of two variables is like a machine with *two* independent dials, and the output depends on both dials' positions simultaneously.

Geometric Interpretation: The graph of $z = f(x, y)$ is a *surface* in three-dimensional space (xyz-space). While a single-variable function gives a curve, a two-variable function gives a surface. For instance, $z = x^2 + y^2$ is the equation of a paraboloid — a bowl-shaped surface.

1.3 Domain and Range

DEFINITION — DOMAIN

The **domain** of $f(x, y)$ is the set of all ordered pairs (x, y) for which the function is defined. It is a region (or set of points) in the xy-plane.

DEFINITION — RANGE

The **range** of $f(x, y)$ is the set of all possible output values z that the function can produce.

Finding the Domain — Rules to Remember:

1. The expression under a **square root** must be ≥ 0.
2. The **denominator** must not be zero.
3. The argument of a **logarithm** must be strictly positive (> 0).
4. The argument of **arcsin / arccos** must lie in $[-1, 1]$.

Example 1.1 — Finding the Domain of a Two-Variable Function

SOLVED

Step 1 — Given Function

Find the domain of `f(x, y) = sqrt(4 − x² − y²)`

Step 2 — Concept Used

> *For a square root to be defined, the expression inside must be greater than or equal to zero. This is because we cannot take the square root of a negative number in the real number system.*

Step 3 — Set Up the Inequality

The expression inside the square root is $4 - x^2 - y^2$.

For the function to be defined, we need:

$$4 - x^2 - y^2 \geq 0$$

Step 4 — Rearrange the Inequality

Move x^2 and y^2 to the right side:

$$x^2 + y^2 \leq 4$$

This means: the sum of squares of x and y must be at most 4.

Step 5 — Geometric Interpretation

The inequality $x^2 + y^2 \leq 4$ describes all points (x, y) at a distance of at most 2 from the origin. This is the interior and boundary of a circle of radius 2 centered at the origin.

Domain: All points (x, y) such that $x^2 + y^2 \leq 4$
This is a closed disk of radius 2 centered at the origin.

1.4 Functions of Three or More Variables

The concept extends naturally. A function of **three variables** is written as $w = f(x, y, z)$, where three independent inputs produce one output. For example, the temperature at a point inside a 3D room depends on x, y, and z.

In general, for *n* variables: $u = f(x_1, x_2, \ldots, x_n)$.

While we cannot visualize these graphs (they live in 4D, 5D, ... space), the mathematical tools of partial differentiation work the same way.

1.5 Notation Summary

Number of Variables	Function Notation	Graph Lives In	Example
One variable	$y = f(x)$	2D plane (xy-plane)	$y = x^2 + 3$
Two variables	$z = f(x, y)$	3D space (xyz-space)	$z = x^2 + y^2$
Three variables	$w = f(x, y, z)$	4D (cannot visualize)	$w = x + y + z$
n variables	$u = f(x_1,\ldots,x_n)$	(n+1)D space	General case

§2 — Continuity and Differentiability

2.1 Limit of a Two-Variable Function

Before we can talk about continuity, we must understand what it means to take a limit of a two-variable function. In single-variable calculus, `x` approaches a point along the number line — there are only two directions: from the left or from the right.

In two-variable calculus, the point `(x, y)` can approach another point `(a, b)` from *infinitely many directions* — along any path in the xy-plane (straight lines at any angle, parabolic curves, spirals, etc.).

DEFINITION — LIMIT OF F(X, Y)

We write:

```
lim[(x,y) → (a,b)] f(x, y) = L
```

if the value of `f(x, y)` gets arbitrarily close to `L` as `(x, y)` approaches `(a, b)` along **every possible path**.

Key Insight: If the function approaches different values along different paths, the limit does NOT exist.

💡 **Important Technique**

To prove a limit EXISTS: use the definition (epsilon-delta) or squeeze theorem.

To prove a limit does NOT EXIST: find two different paths that give two different limit values.

2.2 Continuity of f(x, y)

DEFINITION — CONTINUITY AT A POINT

A function `f(x, y)` is said to be **continuous at the point (a, b)** if all three of the following conditions are satisfied:

1. `f(a, b)` is defined (the function exists at the point).
2. The limit `lim[(x,y)→(a,b)] f(x, y)` exists.
3. The limit equals the function value: `lim[(x,y)→(a,b)] f(x, y) = f(a, b)`

If any ONE of these three conditions fails, the function is **discontinuous** at `(a, b)`.

Geometric Meaning of Continuity: A function is continuous at a point if the surface `z = f(x, y)` has no "holes," "jumps," or "tears" at that point. You can draw the surface near that point without lifting your pen.

2.3 Differentiability of f(x, y)

Differentiability is a stronger condition than continuity. Just as with single-variable functions (where differentiability implies continuity, but not the other way around), the same relationship holds for multivariable functions.

DEFINITION — DIFFERENTIABILITY

A function `f(x, y)` is said to be **differentiable at (a, b)** if there exist numbers A and B (which are the partial derivatives) such that:

$$f(a+h, b+k) - f(a,b) = A \cdot h + B \cdot k + \varepsilon_1 \cdot h + \varepsilon_2 \cdot k$$

where $\varepsilon_1 \to 0$ and $\varepsilon_2 \to 0$ as $(h, k) \to (0, 0)$.

In simple terms: the function can be approximated by a **tangent plane** near `(a, b)`, and the error in this approximation goes to zero faster than the distance to the point.

SUFFICIENT CONDITION FOR DIFFERENTIABILITY

If the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ both exist AND are continuous at `(a, b)`, then `f` is differentiable at `(a, b)`.

Chain of Implications: Continuous partial derivatives ⇒ Differentiable ⇒ Continuous

⚠ Common Exam Trap

Existence of partial derivatives does NOT by itself guarantee differentiability. A function can have both partial derivatives at a point yet still be non-differentiable there. This is a key conceptual point that often appears in exam problems.

§3 — Partial Derivatives — First Order

3.1 The Core Idea — Why Partial Derivatives?

Consider the temperature `T = f(x, y)` on a metal plate. You are standing at point `(x, y)`. You want to know: "If I move only in the x-direction (keeping y fixed), how fast does the temperature change?" This exact question is answered by the partial derivative with respect to x.

Think of it this way: you have a mountain surface `z = f(x, y)`. If you walk only eastward (along x), the slope you feel is the partial derivative with respect to x. If you walk only northward (along y), the slope is the partial derivative with respect to y.

DEFINITION — PARTIAL DERIVATIVE WITH RESPECT TO X

The partial derivative of `f(x, y)` with respect to `x`, written as `∂f/∂x` or f_x, is defined as:

$$\partial f/\partial x = \lim_{h\to 0} [\, f(x+h, y) - f(x, y)\,] / h$$

Key Rule: To compute `∂f/∂x`, differentiate with respect to `x` while treating `y` as a **constant**.

DEFINITION — PARTIAL DERIVATIVE WITH RESPECT TO Y

The partial derivative of `f(x, y)` with respect to `y`, written as `∂f/∂y` or f_y, is defined as:

$$\partial f/\partial y = \lim_{k\to 0} [\, f(x, y+k) - f(x, y)\,] / k$$

Key Rule: To compute `∂f/∂y`, differentiate with respect to `y` while treating `x` as a **constant**.

3.2 Notation Guide

Notation	Meaning	Read as
$\partial f/\partial x$	Partial derivative of f w.r.t. x	"partial f over partial x"
f_x or f_1	Same as above (subscript notation)	"f sub x"
$\partial z/\partial x$	If z = f(x,y), same as $\partial f/\partial x$	"partial z over partial x"
D_1f or D_xf	Operator notation	"D1 of f"

3.3 How to Compute Partial Derivatives — The Method

The mechanical process is beautifully simple once you understand it:

1. To find `∂f/∂x` : **Freeze y** (treat it like any constant such as 2, 5, π, etc.) and differentiate normally with respect to x using all the single-variable rules you know.
2. To find `∂f/∂y` : **Freeze x** (treat it like any constant) and differentiate normally with respect to y.

💡 **Think of it This Way**

When computing `∂/∂x` , mentally replace every `y` in the function with the number 5 (or any constant). Then differentiate as if it were a single-variable function in x. The answer will still have y's in it — that's fine! The y's that remain are just treated as constant "parameters."

Example 3.1 — Partial Derivatives of a Polynomial Function

BASIC

Step 1 — Given Function

Let $f(x, y) = x^3 + 3x^2y + y^3 - 7x + 2y - 10$

Find $\partial f/\partial x$ and $\partial f/\partial y$.

Step 2 — Compute ∂f/∂x (treat y as constant)

> *We freeze y and differentiate term by term with respect to x. For each term, we ask: "how does this term change when x changes, while y stays fixed?"*

Break the function into individual terms:

- $d/dx(x^3) = 3x^2$ (standard power rule)
- $d/dx(3x^2y) = 3y \cdot d/dx(x^2) = 3y \cdot 2x = 6xy$ (y is constant, factor it out)
- $d/dx(y^3) = 0$ (y^3 is purely a constant with no x, so its derivative is zero)
- $d/dx(-7x) = -7$ (standard constant multiple rule)
- $d/dx(2y) = 0$ (2y has no x, so it's a constant — derivative is zero)
- $d/dx(-10) = 0$ (constant — derivative is zero)

Collecting all terms:

$$\partial f/\partial x = 3x^2 + 6xy + 0 + (-7) + 0 + 0 = 3x^2 + 6xy - 7$$

Step 3 — Compute ∂f/∂y (treat x as constant)

> *Now we freeze x and differentiate term by term with respect to y.*

- $d/dy(x^3) = 0$ (x^3 is a constant — no y in it)
- $d/dy(3x^2y) = 3x^2 \cdot d/dy(y) = 3x^2 \cdot 1 = 3x^2$ (x^2 is constant, factor out)
- $d/dy(y^3) = 3y^2$ (standard power rule)
- $d/dy(-7x) = 0$ (-7x is a constant in y)
- $d/dy(2y) = 2$ (standard constant multiple rule)
- $d/dy(-10) = 0$ (constant)

Collecting all terms:

$$\partial f/\partial y = 0 + 3x^2 + 3y^2 + 0 + 2 + 0 = 3x^2 + 3y^2 + 2$$

$$\partial f/\partial x = 3x^2 + 6xy - 7$$

$$\partial f/\partial y = 3x^2 + 3y^2 + 2$$

Example 3.2 — Partial Derivatives Involving Trigonometric Functions

INTERMEDIATE

Step 1 — Given Function

Let $f(x, y) = x^2 \sin(y) + y^2 \cos(x) + e^{xy}$

Find $\partial f/\partial x$ and $\partial f/\partial y$.

Step 2 — Compute ∂f/∂x

> *Treat y as a constant. sin(y) is a constant (like the number sin(30°) = 0.5). cos(x) is a function of x. For e^(xy), use the chain rule where we differentiate e^(xy) with respect to x, and the exponent xy has derivative y (since y is constant).*

- **Term 1:** $x^2 \sin(y)$. Here sin(y) is a constant multiplier. So: $d/dx(x^2 \cdot \sin(y)) = \sin(y) \cdot 2x = 2x \sin(y)$
- **Term 2:** $y^2 \cos(x)$. Here y^2 is a constant multiplier. So: $d/dx(y^2 \cdot \cos(x)) = y^2 \cdot (-\sin(x)) = -y^2 \sin(x)$
- **Term 3:** e^{xy} . Let $u = xy$. Then $du/dx = y$ (since y is constant). By chain rule: $d/dx(e^u) = e^u \cdot du/dx = e^{xy} \cdot y = ye^{xy}$

$$\partial f/\partial x = 2x \sin(y) - y^2 \sin(x) + ye^{xy}$$

Step 3 — Compute ∂f/∂y

> *Now treat x as a constant. cos(x) is a constant. For e^(xy), the exponent xy has derivative x when differentiating with respect to y.*

- **Term 1:** $x^2 \sin(y)$. x^2 is constant. $d/dy(x^2 \sin(y)) = x^2 \cos(y)$
- **Term 2:** $y^2 \cos(x)$. cos(x) is constant. $d/dy(y^2 \cos(x)) = \cos(x) \cdot 2y = 2y \cos(x)$
- **Term 3:** e^{xy} . Let $u = xy$. $du/dy = x$. So: $d/dy(e^{xy}) = e^{xy} \cdot x = xe^{xy}$

$$\partial f/\partial y = x^2 \cos(y) + 2y \cos(x) + xe^{xy}$$

$$\partial f/\partial x = 2x \sin(y) - y^2 \sin(x) + ye^{xy}$$
$$\partial f/\partial y = x^2 \cos(y) + 2y \cos(x) + xe^{xy}$$

Example 3.3 — Partial Derivatives of a Logarithmic Function

INTERMEDIATE

Step 1 — Given Function

Let $f(x, y) = \ln(x^2 + y^2)$

Find $\partial f/\partial x$ and $\partial f/\partial y$.

Step 2 — Recall the Rule for Logarithmic Differentiation

> *If $f(x) = \ln(g(x))$, then $f'(x) = g'(x) / g(x)$. This is the chain rule applied to the natural logarithm. Here, $g(x, y) = x^2 + y^2$.*

Step 3 — Compute ∂f/∂x

Let $u = x^2 + y^2$. So $f = \ln(u)$.

By chain rule: $\partial f/\partial x = (1/u) \cdot \partial u/\partial x$

Now: $\partial u/\partial x = \partial/\partial x(x^2 + y^2) = 2x + 0 = 2x$ (y^2 is constant)

$$\partial f/\partial x = (1/(x^2 + y^2)) \cdot 2x = 2x/(x^2 + y^2)$$

Step 4 — Compute ∂f/∂y

$\partial u/\partial y = \partial/\partial y(x^2 + y^2) = 0 + 2y = 2y$ (x^2 is constant)

$$\partial f/\partial y = (1/(x^2 + y^2)) \cdot 2y = 2y/(x^2 + y^2)$$

Step 5 — Observation

Notice the beautiful symmetry: $\partial f/\partial x = 2x/(x^2+y^2)$ and $\partial f/\partial y = 2y/(x^2+y^2)$. The forms are identical; just replace x with y and vice versa. This makes sense because $f(x,y) = \ln(x^2+y^2)$ is symmetric in x and y.

$$\partial f/\partial x = 2x / (x^2 + y^2) \qquad \partial f/\partial y = 2y / (x^2 + y^2)$$

§4 — Higher Order Partial Derivatives

4.1 Second-Order Partial Derivatives

Just as in single-variable calculus, we can differentiate again after taking the first partial derivative. Since we have two variables x and y, there are **four possible second-order partial derivatives**:

```
1. ∂²f/∂x² = fₓₓ → Differentiate with respect to x TWICE 2. ∂²f/∂y² = fᵧᵧ → Differentiate with respect to y TWICE 3. ∂²f/∂y∂x = fₓᵧ → Differentiate w.r.t x first, then y 4. ∂²f/∂x∂y = fᵧₓ → Differentiate w.r.t y first, then x
```

Items 3 and 4 are called "mixed partial derivatives." There is an extremely important theorem about them:

CLAIRAUT'S THEOREM (SYMMETRY OF MIXED PARTIALS)

If `f(x, y)` has continuous second-order mixed partial derivatives on an open region, then:

$$\partial^2 f/\partial y\partial x = \partial^2 f/\partial x\partial y$$

That is, **the order of differentiation does not matter** (as long as the mixed partials are continuous). This is one of the most beautiful results in multivariable calculus.

💡 **Reading the Notation Carefully**

`∂²f/∂y∂x` means: first differentiate with respect to `x` (the one on the right, closest to f), then differentiate the result with respect to `y`.

Think of it as reading from right to left: `∂/∂y [ ∂f/∂x ]`

Example 4.1 — All Four Second-Order Partial Derivatives

IMPORTANT

Step 1 — Given Function

Let $f(x, y) = x^3y^2 + 2x^2y + 5xy^3 - 3x + 4y$

Find all four second-order partial derivatives and verify Clairaut's Theorem.

Step 2 — Compute First-Order Partials

Computing ∂f/∂x (treat y as constant):

- $d/dx(x^3y^2) = 3x^2y^2$
- $d/dx(2x^2y) = 4xy$
- $d/dx(5xy^3) = 5y^3$
- $d/dx(-3x) = -3$
- $d/dx(4y) = 0$

$$f_x = \partial f/\partial x = 3x^2y^2 + 4xy + 5y^3 - 3$$

Computing ∂f/∂y (treat x as constant):

- $d/dy(x^3y^2) = 2x^3y$
- $d/dy(2x^2y) = 2x^2$
- $d/dy(5xy^3) = 15xy^2$
- $d/dy(-3x) = 0$
- $d/dy(4y) = 4$

$$f_y = \partial f/\partial y = 2x^3y + 2x^2 + 15xy^2 + 4$$

Step 3 — Compute f_{xx} (differentiate f_x w.r.t. x)

Start with: $f_x = 3x^2y^2 + 4xy + 5y^3 - 3$

Differentiate each term with respect to x (y is constant):

- $d/dx(3x^2y^2) = 6xy^2$
- $d/dx(4xy) = 4y$
- $d/dx(5y^3) = 0$
- $d/dx(-3) = 0$

$$f_{xx} = \partial^2 f/\partial x^2 = 6xy^2 + 4y$$

Step 4 — Compute f_{yy} (differentiate f_y w.r.t. y)

Start with: $f_y = 2x^3y + 2x^2 + 15xy^2 + 4$

Differentiate each term with respect to y (x is constant):

- $d/dy(2x^3y) = 2x^3$
- $d/dy(2x^2) = 0$
- $d/dy(15xy^2) = 30xy$
- $d/dy(4) = 0$

$$f_{yy} = \partial^2 f/\partial y^2 = 2x^3 + 30xy$$

Step 5 — Compute f_{xy} (differentiate f_x w.r.t. y)

Start with: $f_x = 3x^2y^2 + 4xy + 5y^3 - 3$

Differentiate each term with respect to y (x is constant):

- $d/dy(3x^2y^2) = 6x^2y$
- $d/dy(4xy) = 4x$
- $d/dy(5y^3) = 15y^2$
- $d/dy(-3) = 0$

$$f_{xy} = \partial^2 f/\partial y\partial x = 6x^2y + 4x + 15y^2$$

Step 6 — Compute f_{yx} (differentiate f_y w.r.t. x)

Start with: $f_y = 2x^3y + 2x^2 + 15xy^2 + 4$

Differentiate each term with respect to x (y is constant):

- $d/dx(2x^3y) = 6x^2y$
- $d/dx(2x^2) = 4x$
- $d/dx(15xy^2) = 15y^2$
- $d/dx(4) = 0$

$$f_{yx} = \partial^2 f/\partial x\partial y = 6x^2y + 4x + 15y^2$$

Step 7 — Verify Clairaut's Theorem

We found: $f_{xy} = 6x^2y + 4x + 15y^2$ and $f_{yx} = 6x^2y + 4x + 15y^2$

Indeed, $f_{xy} = f_{yx}$ ✓ — Clairaut's Theorem is verified!

$f_{xx} = 6xy^2 + 4y$ | $f_{yy} = 2x^3 + 30xy$

$f_{xy} = f_{yx} = 6x^2y + 4x + 15y^2$ (Clairaut's Theorem Verified)

4.2 Euler's Theorem on Homogeneous Functions

DEFINITION — HOMOGENEOUS FUNCTION

A function $f(x, y)$ is said to be **homogeneous of degree n** if:

$$f(tx, ty) = t^n \cdot f(x, y) \text{ for all } t > 0$$

In simpler terms: if you multiply all variables by t, the whole function gets multiplied by t^n, where n is the degree.

Examples:

- $f(x,y) = x^2 + xy + y^2$ is homogeneous of degree 2 (every term has total degree 2)
- $f(x,y) = x^3 + x^2y + y^3$ is homogeneous of degree 3
- $f(x,y) = x + y$ is homogeneous of degree 1
- $f(x,y) = x^2 + xy + y^3$ is NOT homogeneous (terms have degrees 2, 2, and 3)

EULER'S THEOREM FOR HOMOGENEOUS FUNCTIONS

If $f(x, y)$ is a homogeneous function of degree n with continuous first-order partial derivatives, then:

$$x \cdot \partial f/\partial x + y \cdot \partial f/\partial y = n \cdot f(x, y)$$

What this means: The "weighted" sum of partial derivatives (weighted by the variables themselves) equals n times the function.

Example 4.2 — Verify Euler's Theorem

EXAM FAVORITE

Step 1 — Given

Let $f(x, y) = x^3 + 3x^2y + 3xy^2 + y^3$. Verify Euler's Theorem.

Step 2 — Identify Degree

Every term has degree 3 (x^3 has degree 3, x^2y has degree 2+1=3, xy^2 has degree 1+2=3, y^3 has degree 3). So n = 3.

Note: $f(x,y) = (x+y)^3$ is a homogeneous function of degree 3.

Step 3 — Compute ∂f/∂x

$$\partial f/\partial x = 3x^2 + 6xy + 3y^2$$

Step 4 — Compute ∂f/∂y

$$\partial f/\partial y = 3x^2 + 6xy + 3y^2$$

(Note: in this symmetric function, the partials happen to be equal.)

Step 5 — Compute $x \cdot f_x + y \cdot f_y$

$$x \cdot (3x^2 + 6xy + 3y^2) + y \cdot (3x^2 + 6xy + 3y^2)$$

$$= 3x^3 + 6x^2y + 3xy^2 + 3x^2y + 6xy^2 + 3y^3$$

$$= 3x^3 + 9x^2y + 9xy^2 + 3y^3 = 3(x^3 + 3x^2y + 3xy^2 + y^3)$$

$$= 3 \cdot f(x, y)$$

Step 6 — Conclusion

We get `x·fₓ + y·fᵧ = 3f`, which equals `n·f` with n = 3. ✓

Euler's Theorem Verified: x·(∂f/∂x) + y·(∂f/∂y) = 3·f(x,y)

§5 — Total Differential and Total Derivative

5.1 What is the Total Differential?

In single-variable calculus, when $y = f(x)$, the differential is $dy = f'(x)\, dx$. It represents the approximate change in y when x changes by a tiny amount dx.

For a function of two variables $z = f(x, y)$, we want to know: "If both x and y change by small amounts dx and dy, how much does z change?"

DEFINITION — TOTAL DIFFERENTIAL

If $z = f(x, y)$ is a differentiable function, then the **total differential** of z is:

$$dz = (\partial f/\partial x)\, dx + (\partial f/\partial y)\, dy$$

This represents the approximate change in z when x changes by dx and y changes by dy.

Physical Meaning: The change in z comes from two independent contributions — the x-direction change and the y-direction change. These contributions add up.

💡 **Analogy — Error Estimation**

Suppose the area of a rectangle is $A = xy$. If the measurements of x and y have small errors dx and dy respectively, then the error in area is approximately:

$$dA = (\partial A/\partial x)\, dx + (\partial A/\partial y)\, dy = y \cdot dx + x \cdot dy$$

This is exactly how engineers estimate measurement errors in quantities that depend on multiple measured values.

5.2 Total Derivative

Now suppose x and y are not independent — both are functions of a single parameter t. That is, $x = x(t)$ and $y = y(t)$, so $z = f(x(t), y(t))$ is really a function of t alone. The **total derivative** of z with respect to t is:

Total Derivative Formula: $dz/dt = (\partial f/\partial x)(dx/dt) + (\partial f/\partial y)(dy/dt)$ Compact notation: $dz/dt = f_x \cdot x'(t) + f_y \cdot y'(t)$

Interpretation: The total rate of change of z with respect to t has two parts: how z changes due to x changing (times how fast x changes), plus how z changes due to y changing (times how fast y changes).

Example 5.1 — Computing the Total Derivative

IMPORTANT

Step 1 — Given

If $z = x^2y + xy^2$, where $x = \sin(t)$ and $y = \cos(t)$, find dz/dt.

Step 2 — Formula to Use

> *We use the total derivative formula: $dz/dt = (\partial z/\partial x)(dx/dt) + (\partial z/\partial y)(dy/dt)$. This accounts for both channels through which t affects z — through x and through y.*

Step 3 — Find ∂z/∂x (treat y constant)

$$\partial z/\partial x = 2xy + y^2$$

Step 4 — Find ∂z/∂y (treat x constant)

$$\partial z/\partial y = x^2 + 2xy$$

Step 5 — Find dx/dt and dy/dt

$$x = \sin(t) \rightarrow dx/dt = \cos(t)$$

$$y = \cos(t) \rightarrow dy/dt = -\sin(t)$$

Step 6 — Apply the Total Derivative Formula

$$dz/dt = (2xy + y^2) \cdot \cos(t) + (x^2 + 2xy) \cdot (-\sin(t))$$

Now substitute $x = \sin(t)$ and $y = \cos(t)$:

$$dz/dt = (2\sin(t)\cos(t) + \cos^2(t)) \cdot \cos(t) + (\sin^2(t) + 2\sin(t)\cos(t)) \cdot (-\sin(t))$$

$$= 2\sin(t)\cos^2(t) + \cos^3(t) - \sin^3(t) - 2\sin^2(t)\cos(t)$$

$$= \cos^3(t) - \sin^3(t) + 2\sin(t)\cos(t)(\cos(t) - \sin(t))$$

$dz/dt = \cos^3(t) - \sin^3(t) + 2 \sin(t)\cos(t)(\cos(t) - \sin(t))$

Example 5.2 — Error Estimation Using Total Differential

APPLICATION

Step 1 — Given

The area of a rectangle is $A = xy$. If x is measured as 5 m with an error of 0.03 m, and y is measured as 8 m with an error of 0.05 m, find the approximate error in the area.

Step 2 — Identify Known Values

$x = 5$, $y = 8$, $dx = 0.03$, $dy = 0.05$

Step 3 — Compute Partial Derivatives

$$\partial A/\partial x = y = 8 \qquad \partial A/\partial y = x = 5$$

Step 4 — Apply Total Differential Formula

$$dA = (\partial A/\partial x)dx + (\partial A/\partial y)dy = 8 \times 0.03 + 5 \times 0.05$$

$$= 0.24 + 0.25 = 0.49 \text{ m}^2$$

Approximate error in area = 0.49 m^2

§6 — Chain Rule for Multivariable Functions

6.1 Recall — Single Variable Chain Rule

In single-variable calculus, if `y = f(u)` and `u = g(x)`, then:

```
dy/dx = (dy/du) · (du/dx)
```

The idea: if y depends on u, and u depends on x, then how y changes with x is mediated by how u changes with x.

6.2 Chain Rule — Case 1 (One Independent Variable)

If `z = f(x, y)`, where `x = x(t)` and `y = y(t)` (both functions of one variable t), then:

```
dz/dt = (∂z/∂x)(dx/dt) + (∂z/∂y)(dy/dt) Diagram: z / \ x y | | t t
```

The diagram shows two paths from z to t: through x and through y. The total derivative is the sum of contributions from each path.

6.3 Chain Rule — Case 2 (Two Independent Variables)

If `z = f(x, y)`, where `x = x(s, t)` and `y = y(s, t)` (both functions of two variables s and t), then:

```
∂z/∂s = (∂z/∂x)(∂x/∂s) + (∂z/∂y)(∂y/∂s) ∂z/∂t = (∂z/∂x)(∂x/∂t) + (∂z/∂y)(∂y/∂t)
```

Example 6.1 — Chain Rule with Two Independent Variables

IMPORTANT

Step 1 — Given

If $z = x^2 + y^2$, where $x = s \cdot \cos(t)$ and $y = s \cdot \sin(t)$, find $\partial z/\partial s$ and $\partial z/\partial t$.

Step 2 — Compute ∂z/∂x and ∂z/∂y

$$\partial z/\partial x = 2x \qquad \partial z/\partial y = 2y$$

Step 3 — Compute Partial Derivatives of x and y w.r.t. s and t

$$\partial x/\partial s = \cos(t) \qquad \partial x/\partial t = -s \cdot \sin(t)$$

$$\partial y/\partial s = \sin(t) \qquad \partial y/\partial t = s \cdot \cos(t)$$

Step 4 — Apply Chain Rule for ∂z/∂s

$$\partial z/\partial s = (2x)(\cos(t)) + (2y)(\sin(t))$$

Substitute $x = s \cdot \cos(t)$ and $y = s \cdot \sin(t)$:

$$= 2s \cdot \cos(t) \cdot \cos(t) + 2s \cdot \sin(t) \cdot \sin(t)$$

$$= 2s \cdot \cos^2(t) + 2s \cdot \sin^2(t) = 2s(\cos^2 t + \sin^2 t) = 2s \cdot 1 = 2s$$

Step 5 — Apply Chain Rule for ∂z/∂t

$$\partial z/\partial t = (2x)(-s \cdot \sin(t)) + (2y)(s \cdot \cos(t))$$

Substitute $x = s \cdot \cos(t)$ and $y = s \cdot \sin(t)$:

$$= 2s \cdot \cos(t) \cdot (-s \cdot \sin(t)) + 2s \cdot \sin(t) \cdot (s \cdot \cos(t))$$

$$= -2s^2\cos(t)\sin(t) + 2s^2\sin(t)\cos(t) = 0$$

Step 6 — Geometric Insight

Here x = s·cos(t), y = s·sin(t) are **polar coordinates**. And $z = x^2 + y^2 = s^2$, which depends only on s. So ∂z/∂t = 0 makes perfect sense — changing the angle t does not change the distance from origin!

$\partial z/\partial s = 2s \quad \partial z/\partial t = 0$

§7 — Directional Derivative & §8 — Gradient Vector

7.1 Motivation — Slope in an Arbitrary Direction

Partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ give the rate of change of f in the x-direction and y-direction respectively. But what if we want the rate of change in an *arbitrary* direction? This is exactly what the **directional derivative** answers.

Imagine you are standing on a hillside (the surface $z = f(x, y)$). The directional derivative in a given direction tells you how steep the slope is if you walk in that specific direction.

DEFINITION — DIRECTIONAL DERIVATIVE

The **directional derivative** of $f(x, y)$ at a point (a, b) in the direction of a unit vector $\hat{u} = (l, m)$ is:

$$D_{\hat{u}} f(a, b) = l \cdot (\partial f/\partial x) + m \cdot (\partial f/\partial y)$$

where $l = \cos(\alpha)$ and $m = \cos(\beta)$ are the direction cosines of the direction, and $l^2 + m^2 = 1$ (since û is a unit vector).

⚠ **Critical Point:** The direction must ALWAYS be specified as a **unit vector** (a vector of length 1). If you are given a direction vector that is not a unit vector, you MUST normalize it first by dividing by its magnitude.

8.1 The Gradient Vector

DEFINITION — GRADIENT

The **gradient** of $f(x, y)$ is the vector:

$$\text{grad } f = \nabla f = (\partial f/\partial x)\, \hat{\imath} + (\partial f/\partial y)\, \hat{\jmath}$$

In component form: $\nabla f = (f_x, f_y)$

With this notation, the directional derivative becomes simply the dot product:

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}$$

8.2 Maximum Rate of Change

The directional derivative `D_û f = |∇f| cos(θ)`, where θ is the angle between ∇f and $\hat{u}$.

- **Maximum value** of $D_{\hat{u}} f$ is `|∇f|`, achieved when $\hat{u}$ is in the direction of ∇f ($\theta = 0$).
- **Minimum value** of $D_{\hat{u}} f$ is `-|∇f|`, achieved when $\hat{u}$ is opposite to ∇f ($\theta = \pi$).
- **Zero value** when $\hat{u}$ is perpendicular to ∇f ($\theta = \pi/2$).

KEY GEOMETRIC PROPERTIES OF GRADIENT

1. The gradient vector `∇f` points in the direction of **maximum increase** of f.
2. The magnitude `|∇f|` gives the **maximum rate of change**.
3. The gradient is **perpendicular to the level curves** of f (curves where f = constant).

Example 7.1 — Computing Directional Derivative

EXAM FREQUENT

Step 1 — Given

Find the directional derivative of $f(x, y) = x^2 + xy + y^2$ at the point $(1, 2)$ in the direction of the vector $v = (3, 4)$.

Step 2 — Find the Unit Direction Vector

> *The given direction vector v = (3, 4) has magnitude |v| = sqrt(3² + 4²) = sqrt(9 + 16) = sqrt(25) = 5. We divide by 5 to get the unit vector.*

$$|v| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\hat{u} = v/|v| = (3/5, 4/5) \rightarrow l = 3/5,\ m = 4/5$$

Verify: $(3/5)^2 + (4/5)^2 = 9/25 + 16/25 = 25/25 = 1$ ✓

Step 3 — Compute the Gradient (Partial Derivatives)

$$\partial f/\partial x = 2x + y \qquad \partial f/\partial y = x + 2y$$

At the point (1, 2):

$$\partial f/\partial x|_{(1,2)} = 2(1) + 2 = 4$$

$$\partial f/\partial y|_{(1,2)} = 1 + 2(2) = 5$$

So $\nabla f|_{(1,2)} = (4, 5)$

Step 4 — Compute the Directional Derivative

$$D_{\hat{u}} f = l \cdot f_x + m \cdot f_y = (3/5)(4) + (4/5)(5)$$

$$= 12/5 + 20/5 = 32/5 = 6.4$$

Step 5 — Find Maximum Rate of Change

```
|∇f| = sqrt(4² + 5²) = sqrt(16 + 25) = sqrt(41) ≈ 6.4
```

The maximum rate of increase of f at (1,2) is `sqrt(41)`, achieved in the direction of (4, 5).

Directional Derivative D_û f = 32/5 = 6.4
Maximum rate = √41 in direction (4, 5)

§9 — Taylor's & Maclaurin's Expansion (Two Variables)

9.1 Motivation — Why Do We Need Taylor's Expansion?

For a single variable, Taylor's series expands a function near a point as an infinite sum of polynomial terms. This allows us to approximate complex functions (like sin x, cos x, e^x) using simple polynomials. The same idea extends to two variables.

Application: In engineering, complex nonlinear functions are often approximated by linear or quadratic functions near an operating point. Taylor's expansion provides this approximation rigorously.

9.2 Taylor's Expansion for Two Variables

TAYLOR'S SERIES — TWO VARIABLES (EXPANSION ABOUT (A, B))

For a function `f(x, y)`, the Taylor series expansion about the point `(a, b)` is:

$$f(x,y) = f(a,b) + [(x-a)f_x + (y-b)f_y]_{(a,b)}$$

$$+ (1/2!)\ [(x-a)^2 f_{xx} + 2(x-a)(y-b)f_{xy} + (y-b)^2 f_{yy}]_{(a,b)}$$

$$+ (1/3!)\ [(x-a)^3 f_{xxx} + 3(x-a)^2(y-b)f_{xxy} + \ldots]_{(a,b)} + \ldots$$

Using operator notation, this is written compactly as:

$$f(a+h,\ b+k) = f(a,b) + (h\cdot\partial/\partial x + k\cdot\partial/\partial y)f|(a,b)$$

$$+ (1/2!)(h\cdot\partial/\partial x + k\cdot\partial/\partial y)^2 f|(a,b) + \ldots$$

where h = x-a and k = y-b.

10.1 Maclaurin's Series (Expansion about Origin)

MACLAURIN'S SERIES — TWO VARIABLES (EXPANSION ABOUT (0, 0))

Maclaurin's series is Taylor's series with `a = 0, b = 0`. Setting `h = x, k = y`:

$$f(x, y) = f(0,0) + [x \cdot f_x(0,0) + y \cdot f_y(0,0)]$$

$$+ (1/2!)[x^2 \cdot f_{xx}(0,0) + 2xy \cdot f_{xy}(0,0) + y^2 \cdot f_{yy}(0,0)] + \ldots$$

Example 9.1 — Maclaurin's Expansion of e^x · sin(y)

EXAM FAVORITE

Step 1 — Given

Expand $f(x, y) = e^x \cdot \sin(y)$ using Maclaurin's series up to second-degree terms.

Step 2 — Find f(0,0)

$$f(0, 0) = e^0 \cdot \sin(0) = 1 \cdot 0 = 0$$

Step 3 — Compute First-Order Partial Derivatives

$$f_x = e^x \cdot \sin(y) \rightarrow f_x(0,0) = e^0 \cdot \sin(0) = 0$$

$$f_y = e^x \cdot \cos(y) \rightarrow f_y(0,0) = e^0 \cdot \cos(0) = 1 \cdot 1 = 1$$

Step 4 — Compute Second-Order Partial Derivatives

$$f_{xx} = e^x \cdot \sin(y) \rightarrow f_{xx}(0,0) = 0$$

$$f_{yy} = -e^x \cdot \sin(y) \rightarrow f_{yy}(0,0) = 0$$

$$f_{xy} = e^x \cdot \cos(y) \rightarrow f_{xy}(0,0) = 1$$

Step 5 — Write the Maclaurin Expansion

$$f(x,y) = f(0,0) + [x \cdot f_x(0,0) + y \cdot f_y(0,0)] + (1/2!)[x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}] + \ldots$$

$$= 0 + [x \cdot 0 + y \cdot 1] + (1/2)[x^2 \cdot 0 + 2xy \cdot 1 + y^2 \cdot 0] + \ldots$$

```
= y + xy + ...
```

$e^x \sin(y) \approx y + xy + \ldots$ (Maclaurin up to degree 2)

§11 — Jacobians

11.1 What is a Jacobian?

When we change from one set of variables `(x, y)` to another set `(u, v)`, we often need to understand how "areas" or "volumes" transform. The **Jacobian** is a determinant that captures this transformation. It also plays a crucial role in determining whether a set of functions is functionally independent.

Application: In double integrals, when changing from Cartesian (x, y) to polar coordinates (r, θ), the Jacobian tells us the "stretching factor" — the factor by which areas get scaled during the change of variables.

DEFINITION — JACOBIAN OF TWO FUNCTIONS

If `u = u(x, y)` and `v = v(x, y)` are functions of two independent variables x and y, then the **Jacobian of u, v with respect to x, y** is the determinant:

```
J = ∂(u, v)/∂(x, y) = | ∂u/∂x    ∂u/∂y |
```

```
| ∂v/∂x    ∂v/∂y |
```

```
= (∂u/∂x)(∂v/∂y) − (∂u/∂y)(∂v/∂x)
```

11.2 How to Compute a Jacobian — Full Procedure

Step	Action	What to Do
1	Identify functions	Write down u and v clearly
2	Compute 4 partial derivatives	∂u/∂x, ∂u/∂y, ∂v/∂x, ∂v/∂y
3	Set up 2×2 determinant	Top row: partials of u; Bottom row: partials of v
4	Evaluate determinant	(top-left)(bottom-right) - (top-right)(bottom-left)
5	Simplify	Expand and collect terms

Example 11.1 — Jacobian for Polar Coordinate Transformation

CLASSIC EXAM PROBLEM

Step 1 — Given Transformation

Let $x = r\cdot\cos(\theta)$ and $y = r\cdot\sin(\theta)$. Find the Jacobian $\partial(x, y)/\partial(r, \theta)$.

> *This is the polar-to-Cartesian transformation. The Jacobian tells us how areas in r-θ space correspond to areas in x-y space.*

Step 2 — Identify Functions and Variables

Here u = x = r·cos(θ), v = y = r·sin(θ), independent variables are r and θ.

Step 3 — Compute All Four Partial Derivatives

- $\partial x/\partial r = \cos(\theta)$ (differentiate r·cos(θ) w.r.t. r; cos(θ) is constant)
- $\partial x/\partial\theta = -r\cdot\sin(\theta)$ (differentiate r·cos(θ) w.r.t. θ; r is constant)
- $\partial y/\partial r = \sin(\theta)$ (differentiate r·sin(θ) w.r.t. r; sin(θ) is constant)
- $\partial y/\partial\theta = r\cdot\cos(\theta)$ (differentiate r·sin(θ) w.r.t. θ; r is constant)

Step 4 — Set Up the 2×2 Determinant

```
J = ∂(x,y)/∂(r,θ) = | ∂x/∂r    ∂x/∂θ | = | cos(θ)    -r·sin(θ) |
```

```
| ∂y/∂r    ∂y/∂θ | | sin(θ)     r·cos(θ) |
```

Step 5 — Evaluate the Determinant

> *For a 2×2 matrix with elements [a, b; c, d], the determinant = ad - bc. Here: a = cos(θ), b = -r·sin(θ), c = sin(θ), d = r·cos(θ).*

```
J = (cos θ)(r·cos θ) − (−r·sin θ)(sin θ)
```

```
= r·cos²θ + r·sin²θ
```

```
= r(cos²θ + sin²θ)
```

$$= r \cdot 1 = r$$

Using the identity: $\cos^2\theta + \sin^2\theta = 1$

Step 6 — Interpretation

The Jacobian J = r means that when we change from polar to Cartesian coordinates in a double integral, we multiply by r: $dx \cdot dy = r \cdot dr \cdot d\theta$. This is why the area element in polar form is r·dr·dθ, which you have probably seen in integration.

$J = \partial(x, y)/\partial(r, \theta) = r$

📚 Example 11.2 — Jacobian of u = x + y, v = x - y

STRAIGHTFORWARD

Step 1 — Given

If $u = x + y$ and $v = x - y$, find $\partial(u, v)/\partial(x, y)$.

Step 2 — Compute Partial Derivatives

- $\partial u/\partial x = 1$ (differentiate x+y w.r.t. x)
- $\partial u/\partial y = 1$ (differentiate x+y w.r.t. y)
- $\partial v/\partial x = 1$ (differentiate x-y w.r.t. x)
- $\partial v/\partial y = -1$ (differentiate x-y w.r.t. y)

Step 3 — Evaluate Determinant

```
J = | 1   1 | = (1)(-1) - (1)(1) = -1 - 1 = -2
```

```
| 1 -1 |
```

$J = \partial(u, v)/\partial(x, y) = -2$

11.3 Properties of Jacobians

KEY PROPERTIES

1. **Chain Rule for Jacobians:** If `(u, v)` are functions of `(r, s)`, and `(r, s)` are functions of `(x, y)`, then: `∂(u,v)/∂(x,y) = [∂(u,v)/∂(r,s)] × [∂(r,s)/∂(x,y)]`
2. **Inverse Transformation:** `∂(x,y)/∂(u,v) = 1 / ∂(u,v)/∂(x,y)` (the Jacobians are reciprocals)
3. **Functional Dependence:** If `J = 0`, then u and v are functionally dependent (one can be expressed as a function of the other).

§12 — Functional Dependence

DEFINITION — FUNCTIONAL DEPENDENCE

Two functions `u = u(x, y)` and `v = v(x, y)` are said to be **functionally dependent** if there exists a function F such that `F(u, v) = 0`, meaning v can be expressed purely in terms of u (or vice versa) — without any direct need for x or y.

Test: u and v are functionally dependent if and only if their Jacobian is identically zero:

```
J = ∂(u, v)/∂(x, y) = 0    ⟺    u and v are functionally dependent
```

Example 12.1 — Testing Functional Dependence

EXAM FREQUENT

Step 1 — Given

Show that $u = x + y$ and $v = x^2 + 2xy + y^2$ are functionally dependent.

Step 2 — Compute Partial Derivatives

$$\partial u/\partial x = 1, \quad \partial u/\partial y = 1$$

$$\partial v/\partial x = 2x + 2y, \quad \partial v/\partial y = 2x + 2y$$

Step 3 — Compute Jacobian

$$J = \begin{vmatrix} 1 & 1 \\ 2x+2y & 2x+2y \end{vmatrix}$$

$$= 1\cdot(2x+2y) - 1\cdot(2x+2y) = 0$$

Step 4 — Conclusion and Verification

Since J = 0, u and v are functionally dependent. Let's verify directly:

$$v = x^2 + 2xy + y^2 = (x + y)^2 = u^2$$

So $v = u^2$ — confirming that v is purely a function of u.

$J = 0 \rightarrow$ u and v are functionally dependent; in fact $v = u^2$

§13 — Maxima and Minima of Two Variables

13.1 Motivation — Finding Optimal Points on a Surface

In single-variable calculus, we find the maximum or minimum of a function `f(x)` by finding where `f'(x) = 0`. For functions of two variables, we ask: at what point `(x, y)` does the surface `z = f(x, y)` reach its highest peak (maximum) or its lowest valley (minimum)?

DEFINITION — LOCAL MAXIMUM AND MINIMUM

A point `(a, b)` is called a **local maximum** of f if `f(a, b) ≥ f(x, y)` for all (x, y) in some small neighborhood of (a, b).

A point `(a, b)` is called a **local minimum** of f if `f(a, b) ≤ f(x, y)` for all (x, y) in some small neighborhood of (a, b).

13.2 Stationary (Critical) Points

A necessary condition for a local extremum is that both partial derivatives equal zero simultaneously:

```
Necessary Condition for Extremum: ∂f/∂x = 0 AND ∂f/∂y = 0 simultaneously These equations give the "stationary points" or "critical points" of f.
```

13.3 The Second-Derivative Test (Hessian Test)

Finding a stationary point is not enough — it could be a maximum, a minimum, or a saddle point. To determine the nature, we use the second-order partial derivatives:

SECOND DERIVATIVE TEST — NATURE OF STATIONARY POINTS

Let `(a, b)` be a stationary point. Define:

$$A = f_{xx}(a,b), \quad B = f_{xy}(a,b), \quad C = f_{yy}(a,b)$$

$$D = AC - B^2 \quad \text{(the discriminant)}$$

Condition	Nature of Point (a, b)

$D > 0$ and $A > 0$	Local Minimum
$D > 0$ and $A < 0$	Local Maximum
$D < 0$	Saddle Point (neither max nor min)
$D = 0$	Test is inconclusive (further investigation needed)

13.4 Step-by-Step Procedure for Finding Maxima/Minima

Step	Action
1	Compute f_x and f_y
2	Set $f_x = 0$ and $f_y = 0$ simultaneously and solve for (x, y)
3	For each critical point, compute $A = f_{xx}$, $B = f_{xy}$, $C = f_{yy}$
4	Calculate $D = AC - B^2$
5	Apply the second-derivative test to classify each point
6	Find the function value at the extremum points

Example 13.1 — Finding Maxima/Minima and Saddle Points

ULTRA IMPORTANT

Step 1 — Given Function

Find the maxima, minima, and saddle points of $f(x, y) = x^3 + y^3 - 3x - 3y + 4$

Step 2 — Compute First-Order Partial Derivatives

$$f_x = \partial f/\partial x = 3x^2 - 3$$

$$f_y = \partial f/\partial y = 3y^2 - 3$$

Step 3 — Set Both Equal to Zero and Solve

Setting $f_x = 0$:

$$3x^2 - 3 = 0 \rightarrow 3x^2 = 3 \rightarrow x^2 = 1 \rightarrow x = \pm 1$$

Setting $f_y = 0$:

$$3y^2 - 3 = 0 \rightarrow 3y^2 = 3 \rightarrow y^2 = 1 \rightarrow y = \pm 1$$

The stationary points are all combinations: $(1,1),\ (1,-1),\ (-1,1),\ (-1,-1)$

Step 4 — Compute Second-Order Partial Derivatives

$$f_{xx} = 6x \qquad f_{yy} = 6y \qquad f_{xy} = 0$$

Step 5 — Classify Each Critical Point

Point (1, 1):

$$A = f_{xx}(1,1) = 6(1) = 6, \quad B = f_{xy}(1,1) = 0, \quad C = f_{yy}(1,1) = 6(1) = 6$$

$$D = AC - B^2 = 6\cdot 6 - 0^2 = 36 > 0 \text{ and } A = 6 > 0 \rightarrow \text{LOCAL MINIMUM}$$

$$f(1,1) = 1 + 1 - 3 - 3 + 4 = 0$$

Point (-1, -1):

$$A = f_{xx}(-1,-1) = 6(-1) = -6, \quad B = 0, \quad C = f_{yy}(-1,-1) = 6(-1) = -6$$

$$D = (-6)(-6) - 0^2 = 36 > 0 \text{ and } A = -6 < 0 \rightarrow \text{LOCAL MAXIMUM}$$

$$f(-1,-1) = -1 - 1 + 3 + 3 + 4 = 8$$

Point (1, -1):

$$A = f_{xx}(1,-1) = 6(1) = 6, \quad B = 0, \quad C = f_{yy}(1,-1) = 6(-1) = -6$$

$$D = (6)(-6) - 0^2 = -36 < 0 \rightarrow \text{SADDLE POINT}$$

Point (-1, 1):

$$A = f_{xx}(-1,1) = 6(-1) = -6, \quad B = 0, \quad C = f_{yy}(-1,1) = 6(1) = 6$$

$$D = (-6)(6) - 0^2 = -36 < 0 \rightarrow \text{SADDLE POINT}$$

Local Maximum: (-1, -1) with f(-1,-1) = 8
Local Minimum: (1, 1) with f(1,1) = 0
Saddle Points: (1,-1) and (-1,1)

§14 — Lagrange's Method of Undetermined Multipliers

14.1 The Problem — Constrained Optimization

In Section 13, we found maxima and minima with no restrictions on x and y (unconstrained). But in many real engineering problems, we optimize a function subject to one or more *constraints*. For example:

- Find the dimensions of the box with maximum volume given a fixed surface area.
- Find the point on the plane $x + y + z = 1$ closest to the origin.
- Maximize profit given a budget constraint.

THE LAGRANGE SETUP

We want to find the extremum of `f(x, y)` (called the **objective function**) subject to the constraint `g(x, y) = 0` (called the **constraint function**).

LAGRANGE'S THEOREM — METHOD OF MULTIPLIERS

If `f(x, y)` has an extremum at `(a, b)` subject to the constraint `g(x, y) = 0`, then there exists a scalar `λ` (called the **Lagrange multiplier**) such that:

$$\partial f/\partial x = \lambda \cdot \partial g/\partial x \quad \text{and} \quad \partial f/\partial y = \lambda \cdot \partial g/\partial y$$

Combined with the constraint `g(x, y) = 0`, we get a system of three equations in three unknowns `(x, y, λ)`.

14.2 Step-by-Step Procedure

Step	What to Do
1	Identify objective $f(x,y)$ and constraint $g(x,y) = 0$
2	Write the Lagrange conditions: $f_x = \lambda g_x$, $f_y = \lambda g_y$
3	Add the constraint equation: $g(x,y) = 0$
4	Solve the system of three equations for x, y, and λ
5	Evaluate f at the solution point — this is the extremum

6	Determine whether it is a maximum or minimum (by context or comparison)

Example 14.1 — Maximize Area of Rectangle with Fixed Perimeter

CLASSIC APPLICATION

Step 1 — Problem Setup

A rectangle has perimeter 20 units. Find the dimensions that maximize its area.

Let x = length and y = width.

Objective: Maximize $f(x, y) = xy$ (area)

Constraint: $2x + 2y = 20$, i.e., $g(x, y) = x + y - 10 = 0$

Step 2 — Write Lagrange Conditions

$$f_x = \lambda \cdot g_x \rightarrow y = \lambda \cdot 1 = \lambda \quad \ldots \text{(i)}$$

$$f_y = \lambda \cdot g_y \rightarrow x = \lambda \cdot 1 = \lambda \quad \ldots \text{(ii)}$$

$$\text{Constraint: } x + y = 10 \quad \ldots \text{(iii)}$$

(where $f_x = y$, $f_y = x$, $g_x = 1$, $g_y = 1$)

Step 3 — Solve the System

From (i): $\lambda = y$

From (ii): $\lambda = x$

Therefore: $x = y$

Substituting into (iii): $x + x = 10 \rightarrow 2x = 10 \rightarrow x = 5$

So $y = 5$.

Step 4 — Compute Maximum Area

$$f(5, 5) = 5 \times 5 = 25 \text{ square units}$$

This tells us that among all rectangles with perimeter 20, the one with maximum area is the **square** (x = y = 5). This is a classic geometric result confirmed by Lagrange multipliers.

Maximum Area = 25 sq. units at x = y = 5 units (a square)

Example 14.2 — Lagrange Multiplier with Three Variables

ADVANCED

Step 1 — Problem Setup

Find the minimum value of $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + y + z = 1$.

(Geometrically: Find the point on the plane x+y+z=1 closest to the origin.)

Objective: Minimize $f = x^2 + y^2 + z^2$

Constraint: $g = x + y + z - 1 = 0$

Step 2 — Lagrange Conditions (3 variables)

$$f_x = \lambda g_x: \quad 2x = \lambda \quad \ldots(i)$$

$$f_y = \lambda g_y: \quad 2y = \lambda \quad \ldots(ii)$$

$$f_z = \lambda g_z: \quad 2z = \lambda \quad \ldots(iii)$$

$$\text{Constraint:} \quad x + y + z = 1 \quad \ldots(iv)$$

Step 3 — Solve

From (i), (ii), (iii): $2x = 2y = 2z = \lambda$, so $x = y = z$

From (iv): $x + x + x = 1 \rightarrow 3x = 1 \rightarrow x = 1/3$

So $x = y = z = 1/3$.

Step 4 — Compute Minimum Value

$$f(1/3, 1/3, 1/3) = (1/3)^2 + (1/3)^2 + (1/3)^2 = 1/9 + 1/9 + 1/9 = 3/9 = 1/3$$

Minimum value = 1/3 at (1/3, 1/3, 1/3)